Letter of Motivation

Md. Modacher Mahmud Rafi
Student ID: 201-35-584
B.Sc. in Software Engineering (Major in Data Science)
Daffodil International University

Faculty of Computer Science and Mathematics
M.Sc. Artificial Intelligence Engineering
University of Passau
Innstraße 41, 94032 Passau

Dear Members of the Admissions Authority,

I am writing to apply for admission to the M.Sc. in Artificial Intelligence Engineering at the University of Passau, commencing in [Winter Semester 2026 / Summer Semester 2027]. Over the past four years, my Bachelor of Science in Software Engineering with a major in Data Science at Daffodil International University has progressively focused my interests at the intersection of mathematics, software engineering, and machine learning. The Passau M.Sc. AI Engineering programme — with its research-oriented curriculum, its emphasis on AI System Engineering, and its explicit treatment of cross-cutting legal, ethical and societal concerns — is precisely the next step I want to take in my academic and professional development.

My undergraduate programme combined a strong computing core with applied data science. On the mathematical and theoretical side, I completed Mathematics I and II (calculus, linear algebra and Fourier analysis), Discrete Mathematics, Statistics, Theory of Computing and Algorithm Design and Analysis, building the formal foundation that the Algorithm Engineering and Mathematical Modelling module group at Passau builds on. On the computer science side, I worked through Data Structures, Object-Oriented Programming and Design, Database Systems, Operating Systems and Systems Programming, Computer Architecture, Software Architecture and Design, Software Quality Assurance and Testing, Software Project Management, and Compiler Design. I graduated with a CGPA of 3.61 out of 4.00 over 145 credits, with strong performance in the courses most relevant to AI: Artificial Intelligence (A+), Introduction to Machine Learning (A+), and Algorithms Design and Analysis (A+).

Within my Data Science major I worked through Statistical Data Analysis, Machine Learning Driven Data Analysis I and II, and a Data Science Capstone Project, gaining hands-on experience with the full ML workflow from data engineering to model evaluation and the communication of results. My undergraduate thesis, "Machine Learning Approach to Predict Flood in Bangladesh from Historical Data", was an individual research project conducted under the supervision of Mr. Khalid Been Badruzzaman Biplob (Lecturer, Department of Software Engineering). I designed and executed an end-to-end ML pipeline on 21,120 meteorological records spanning 34 stations from 1948 to 2013: data cleaning and imputation, feature engineering, label encoding and standardisation, an 80/20 train-test split, and the implementation and comparison of six regression models (Polynomial Regression, Random Forest, K-Nearest Neighbours, Gradient Boosting, XGBoost and an MLP Regressor). KNN achieved the strongest test-set R^2 of 0.90, with Random Forest (0.85) and XGBoost (0.82) close behind. The work taught me not only modelling and evaluation, but also the harder lesson — that real-world AI systems live or die by the quality of their data engineering and the appropriateness of the chosen evaluation method for the deployment context.

Passau's curriculum maps directly onto the directions I want to grow in. The Artificial Intelligence Methods group would let me move beyond the classical regression and ensemble techniques I used in my thesis into reinforcement learning, knowledge representation and the theoretical foundations of learning systems. The AI System Engineering module group is, in many ways, the most exciting to me: my software engineering background has shown me how often promising models fail in production because of missing testing, evaluation, traceability or operationalisation strategies, and I want formal training in exactly those areas. The AI Applications group resonates with my flood prediction work and with my interest in deploying AI in domains where the stakes — environmental, public-safety, economic — are real. Finally, the Cross-Cutting Concerns group aligns with my conviction, formed while building flood prediction models for a country that is on the front line of climate change, that legal, ethical and societal reasoning cannot be a postscript to AI engineering; it has to be designed in from the start.

Alongside my studies, I have been working in marketing and applied AI at GenAI Labs, a MENA-focused agency that builds custom AI agents, voice bots and analytics systems for businesses. The role has given me a practical, business-side view of what it takes to take AI from prototype to production in a regulated and culturally diverse market — including data governance, evaluation against real customer needs, and clear communication of model behaviour to non-technical stakeholders. This experience has reinforced, rather than replaced, my desire to deepen my technical foundations. The AI products I want to help build over the next decade demand a level of rigour in algorithms, system design and ethical reasoning that I can only acquire through a research-oriented master's programme.

I am drawn to Passau in particular for several reasons. The programme is fully taught in English and is structured around 120 ECTS over four semesters with a substantial master's thesis, giving me both the breadth across module groups and the depth of an independent research contribution. The faculty's superb staff-to-student ratio and small learning groups will allow me to engage closely with researchers and peers — something I valued in my undergraduate thesis supervision and want to continue. Passau's strong industry partnerships and its location within Bavaria's research and engineering ecosystem also offer concrete bridges from coursework to applied research and internships.

My medium-term goal is to work as an AI engineer building AI-based systems where reliability, explainability and societal impact are first-class design concerns — particularly in climate, environmental and public-sector applications. In the longer term, I see a path toward doctoral research in trustworthy and operational AI. The M.Sc. Artificial Intelligence Engineering at Passau is, I believe, the most coherent programme I have found for that trajectory: rigorous in its mathematical and algorithmic foundations, structured around real engineering practice, and explicitly engaged with the cross-cutting questions that AI now forces every engineer to confront.

I would be honoured to be considered for admission and to contribute to Passau's research community. Thank you for taking the time to review my application.

Yours sincerely,

Md. Modacher Mahmud Rafi